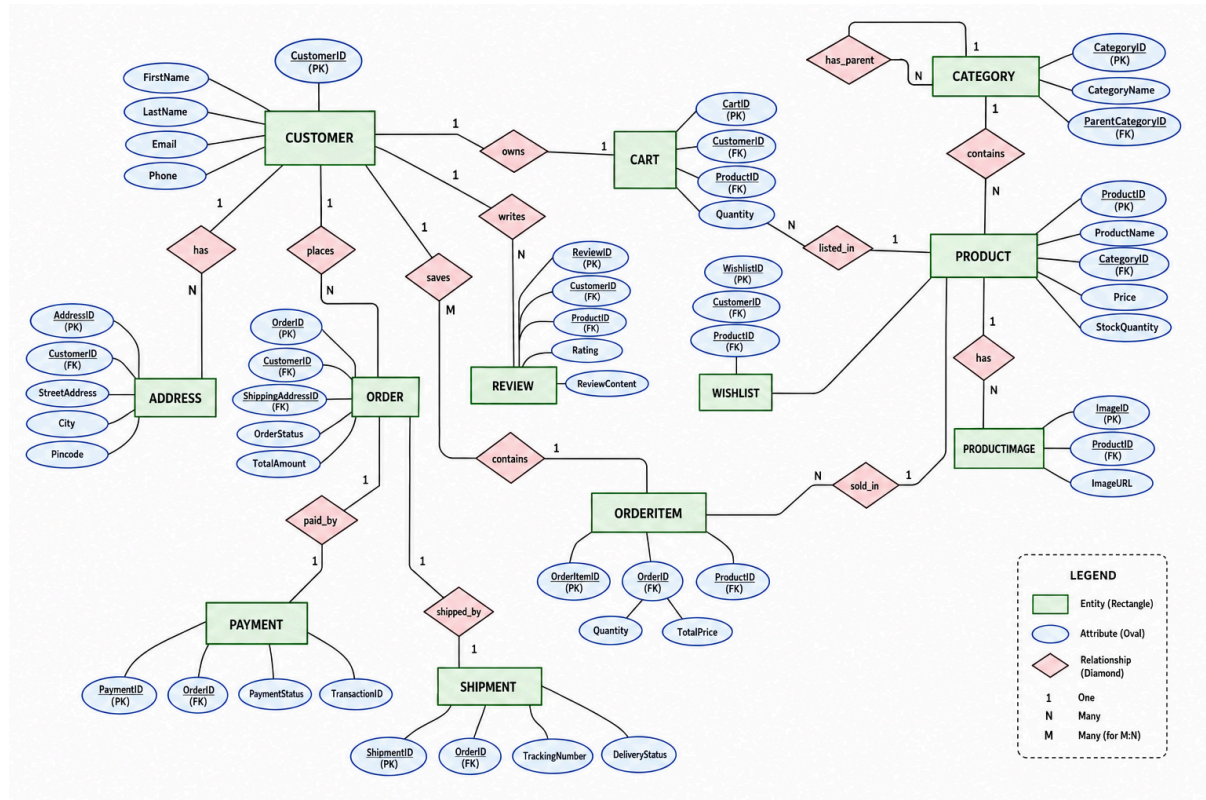


Entity-Relationship (ER) diagram for an e-commerce platform

Diagram : 👍



1. Overview of Core Entities & Modules

The entities are grouped into functional modules:

A. User Management

- **CUSTOMER**: Stores account information like **CustomerID**, name, email, phone, DOB, and **AccountStatus**.
- **ADDRESS**: Stores delivery addresses associated with a customer (**AddressID**, **CustomerID**, street, city, pincode).
 - **Relationship**: A customer can have multiple addresses.

B. Product Catalog & Organization

- **PRODUCT**: Contains item details (**ProductID**, **ProductName**, **Price**, **StockQuantity**, and **CategoryID**).
- **CATEGORY**: Organizes products (**CategoryID**, **CategoryName**).
 - **Self-Reference**: Contains **ParentCategoryID** pointing back to **CATEGORY** to allow hierarchical subcategories (e.g., *Electronics* → *Laptops*).
- **PRODUCTIMAGE**: Holds multiple image URLs for a single product (**ImageID**, **ProductID**, **ImageURL**).

C. Shopping & Customer Engagement

- **CART**: Tracks temporary items selected for purchase (**CartID**, **CustomerID**, **ProductID**, **Quantity**).
- **WISHLIST**: Allows customers to save items for later (**WishlistID**, **CustomerID**, **ProductID**).
- **REVIEW**: Captures customer feedback and ratings (**ReviewID**, **CustomerID**, **ProductID**, **Rating**, **ReviewContent**).

D. Order Processing, Payment & Fulfillment

- **ORDER**: Represents a completed purchase order (**OrderID**, **CustomerID**, **ShippingAddressID**, **OrderStatus**, **TotalAmount**).
- **ORDERITEM**: A junction entity breaking down individual items inside an order (**OrderItemID**, **OrderID**, **ProductID**, **Quantity**, **TotalPrice**).
- **PAYMENT**: Tracks transaction details (**PaymentID**, **OrderID**, **PaymentStatus**, **TransactionID**).
- **SHIPMENT**: Handles physical fulfillment (**ShipmentID**, **OrderID**, **TrackingNumber**, **DeliveryStatus**).

2. Key Relationships & Business Logic

1. **Hierarchy & Categorization**:
 - A **CATEGORY** **contains** multiple **PRODUCTS**.
 - A **CATEGORY** can have a **parent category** (self-referential).
2. **Order Lifecycle**:
 - A **CUSTOMER** **places** an **ORDER**.
 - An **ORDER** **has** multiple **ORDERITEM** lines (linking products to orders with specific quantities/prices).
 - An **ORDER** is **paid by** a **PAYMENT** transaction.
 - An **ORDER** is **shipped to** an **ADDRESS** and tracked via a **SHIPMENT**.
3. **Customer Interaction**:
 - A **CUSTOMER** **owns** a **CART** and a **WISHLIST**.
 - A **CUSTOMER** **writes** a **REVIEW** for a **PRODUCT**.

Rectangles: Entities (e.g., Customer, Product, Order).

Ovals: Attributes (e.g., CustomerID, Name, Price, DOB, Email).

Diamonds: Relationships (e.g., places, has, contains, paid_by).

Lines: Connect them with 1 or N/M labels to show cardinality.